

Experience

Apple • Senior Software Engineer (Contract)

System Intelligence Team, Machine Learning and AI Group March 2024 – Sep 2025

- **ML optimization software:** Transformed research prototype Talaria into production system enabling Apple foundation model latency optimization using Flask, Vue, Svelte, and K8s, improving on-device inference efficiency of 50+ machine learning vision models
- **Designed and built an LLM platform** with internal Next.js storefronts for configuring, integrating, evaluating, and sharing LLMs across teams
- **Open-source visualization:** Expanded interactive tool Mycelium (open-source), a TypeScript and Svelte library used for visualizing deep neural networks and Transformer architectures
- **Performance optimization:** Implemented WebGL viewer achieving 5x higher framerates (120+) when viewing LLMs, and enhanced Mycelium’s feature set for models running on Apple Neural Engine
- **Advanced research workflow** by engineering Python toolchain enabling researchers to compile models to hardware with high reliability (≥ 99.9%)

Memorial Sloan Kettering Cancer Center • Senior Software Engineer

Pathology Results Team, Clinical Bioinformatics Group May 2018 – Present

- **Tech Lead for MPath**, a clinical software for displaying genomic testing results and matching patients to targeted therapies. Leading 8 developers while establishing roadmaps and collaborations across 5 teams.
- **Enterprise healthcare platform:** Architected and implemented performant web apps for managing and visualizing high-throughput clinical data supporting 120k+ patients, using React, Flask, MySQL, and MongoDB. Domain expert and code steward for all front-end releases.
- **Delivered robust front-end performance:** Redesigned and rewrote core UIs, improving query times, performance metrics, code splitting, end-to-end testing, state machines, and application monitoring, achieving 24% faster clinical report turnaround times
- **Spearheaded development lifecycle** by creating build automations, design system, and CI/CD pipelines for 40+ developers to achieve code optimization, visual unity, and standardized tooling across 10+ portals
- **Scalable data pipelines:** Developed large-scale ETL systems and APIs delivering 2x faster processing times using Python, Airflow, and K8s

Skills

Languages • JavaScript, TypeScript, Python, Ruby, SQL, HTML, CSS, Sass

Libraries • React, Flask, FastAPI, Next.js, Vue, Tailwind, Node.js, Playwright, Cypress, pandas

Tools • Git, Docker, GitLab CI/CD, GitHub Actions, Apache Airflow, nginx, AWS S3, ELK, K8s

Design • Figma, Adobe Photoshop, Lightroom, Illustrator

Education

Macaulay Honors College New York, NY

B.A. Chemistry and Bioinformatics, Minor in Mathematics, GPA 3.95 Aug 2013 – May 2017

- Data Science Fellowship 2016
- Full Merit Scholarship 2013 – 2017

Publications

Khwaja E, Song YS, **Agarunov A**, Huang B. CELL-E 2: Translating Proteins to Pictures and Back with a Bidirectional Text-to-Image Transformer. NeurIPS (2023).

Brannon AR, [et al, including **Agarunov A**]. Enhanced specificity of clinical high-sensitivity tumor mutation profiling in cfDNA via paired normal sequencing using MSK-ACCESS. **Nature Communications** (2021).

Projects

Turbokit • Framework for building production-ready, scalable, monorepo microfrontend React apps

ronivonu • Generative art and creative coding projects created in JavaScript, Processing, and Python